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FINANCE FUNDAMENTALS · FORMULA GUIDE

The Math of Time Value of Money

Every formula for moving money across time: growing it forward, pulling it back to today, adding up a stream of payments, and the shortcuts for payments that follow a pattern. Each comes with a plain-English meaning, a worked example, and the mistakes to avoid. All examples use made-up numbers.

A quick note: I'm not a finance professional. I'm sharing what I've learned, in plain language, so it's easier for the next person. Nothing here is financial advice.

How the pieces fit: put every amount on a timeline (part 1) → grow one amount forward (part 2) or pull it back to today (part 3) → move any amount between any two dates (part 4) → add up a stream of different amounts (part 5) → use shortcuts when the payments follow a pattern (part 6) → combine pieces for anything messier (part 7).

Symbols used throughout: C = a cash amount, r = interest rate per period (as a decimal: 5% = 0.05), n or N = number of periods or payments, g = growth rate per period, t = a point in time. Period 0 is today.

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1. Timelines: draw it before you calculate

When does each amount happen, and in which direction does the money go?

Money at different dates can't be compared or added directly. A dollar in year 4 is not the same thing as a dollar today. So the first step is always a timeline: mark period 0 as today, put each amount at the end of the period it arrives, show money you receive as positive and money you pay as negative, and write the rate between the ticks.

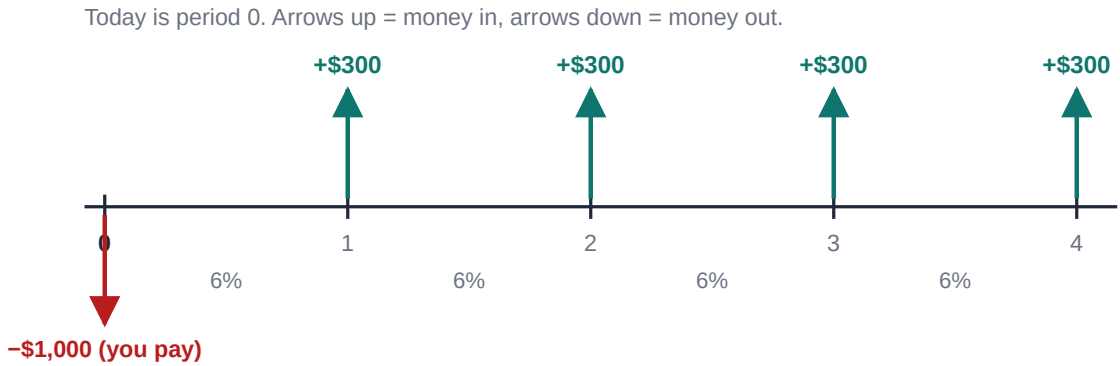


Figure 1. A made-up deal on a timeline: pay $1,000 today, get $300 at the end of each of the next four years, with 6% per period written between the ticks.

Watch out

- "End of year 1" and "start of year 2" are the same point on the timeline.
- Period length must match the rate. A yearly rate goes with yearly periods; monthly payments need a monthly rate.
- Only add amounts once they sit at the same date.

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2. Future value of one amount

If I put money away today, how much will it grow to?

$$\mathrm{FV} = C \times (1 + r)^{n}$$

Each period the amount grows by the rate, and the next period's growth is calculated on the new, bigger total. That is **compounding**: you earn interest on your interest. The part of the growth that comes only from earlier interest is the **interest on interest**.

$$\text{Interest on interest} = \text{total interest} - C \times r \times n$$

Example. $1,000 at 5% a year for 3 years.

Year 1	$1,000 × 1.05 = **$1,050.00**
Year 2	$1,050.00 × 1.05 = **$1,102.50**
Year 3	$1,102.50 × 1.05 = **$1,157.63** (same as $1,000 × 1.05^3)
Total interest	$1,157.63 − $1,000 = **$157.63**
Simple interest	$1,000 × 0.05 × 3 = **$150.00**
Interest on interest	$157.63 − $150.00 = **$7.63**

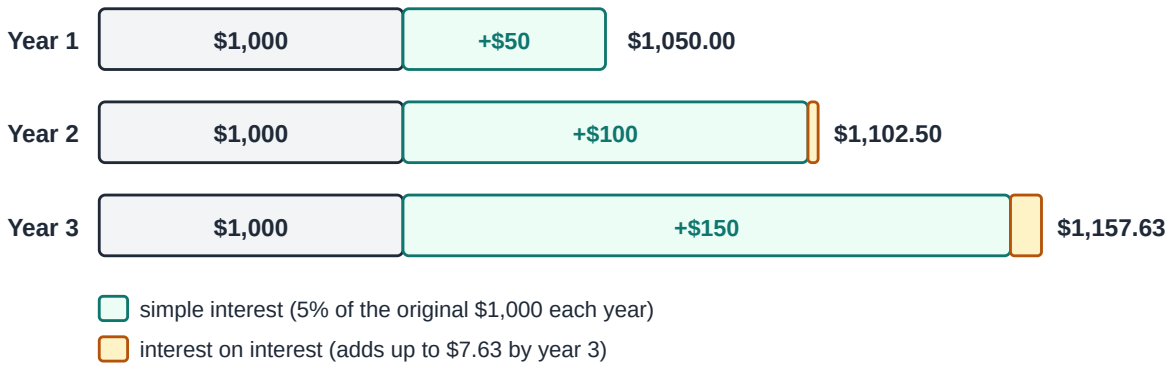


Figure 2. $1,000 at 5% a year. The original amount earns $50 a year; the small amber sliver is interest earned on earlier interest, and it grows every year.

Watch out

- The exponent is the number of periods, not the rate.
- Interest on interest looks tiny over a few years but becomes most of the growth over long ones. That is why time matters so much.
- Use the rate as a decimal: 5% is 0.05, so the growth factor is 1.05, not 6.

3. Present value of one amount

Money arrives later. What is it worth to me today?

$$\mathrm{PV} = \frac{C}{(1+r)^{n}}$$

Present value runs compounding backwards, which is called **discounting**. Instead of multiplying by (1 + r) for each period, you divide by it. A future amount is worth less today because money you have now could be earning a return in the meantime.

Example. You will receive $1,000 in 3 years and the rate is 5%.

One step at a time	$1,000 ÷ 1.05 = $952.38 (year 2) → ÷ 1.05 = $907.03 (year 1) → ÷ 1.05 = **$863.84** (today)
All at once	$1,000 ÷ 1.05^3 = $1,000 ÷ 1.157625 = **$863.84**
Check	$863.84 × 1.05^3 = $1,000.00. Present value and future value undo each other.

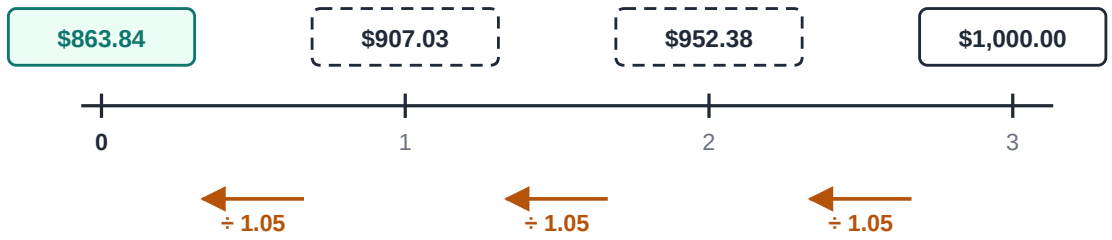


Figure 3. Discounting walks a future amount back one period at a time. $1,000 three years out is worth $863.84 today at 5%.

Watch out

- A higher rate or a longer wait means a lower present value.
- Don't subtract the interest (that would give $1,000 − $150 = $850). Divide by the growth factor instead.

4. Moving any amount between any two dates

The amount isn't today and I don't need it today. How do I move it from one date to another?

$$V_{\text{later}} = V_{\text{earlier}} \times (1 + r)^{\text{gap}}$$

$$V_{\text{earlier}} = \frac{V_{\text{later}}}{(1 + r)^{\text{gap}}}$$

Parts 2 and 3 are two versions of one rule. The "gap" is the number of periods between the two dates. Moving forward in time, multiply; moving backward, divide.

Example. $500 arrives at the end of year 2, and the rate is 5%.

Move it to year 5	gap = 5 − 2 = 3 → $500 × 1.05³ = **$578.81**
Move it to today	gap = 2 − 0 = 2 → $500 ÷ 1.05² = **$453.51**

Watch out

- Count the gap between the two dates, not the date you're moving to.
- This is the move you'll use most in parts 6 and 7, when a shortcut gives a value on the wrong date.

5. A stream of different amounts

I get several payments of different sizes at different times. What are they worth together?

$$\text{PV} = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \cdots + \frac{C_N}{(1+r)^N} = \sum_{t=1}^{N} \frac{C_t}{(1+r)^t}$$

There's no shortcut for uneven amounts. Bring each one back to today on its own (part 3), then add them up. Once they all sit at period 0 they can be added.

Example. $100 at year 1, $200 at year 2, $300 at year 3, at 5%.

Year 1	$100 ÷ 1.05 = $95.24
Year 2	$200 ÷ 1.05² = $181.41
Year 3	$300 ÷ 1.05³ = $259.15
Value today	**$535.80**
Value at year 3 instead	$535.80 × 1.05³ = **$620.25** (any date works, as long as every amount is moved to the same one)

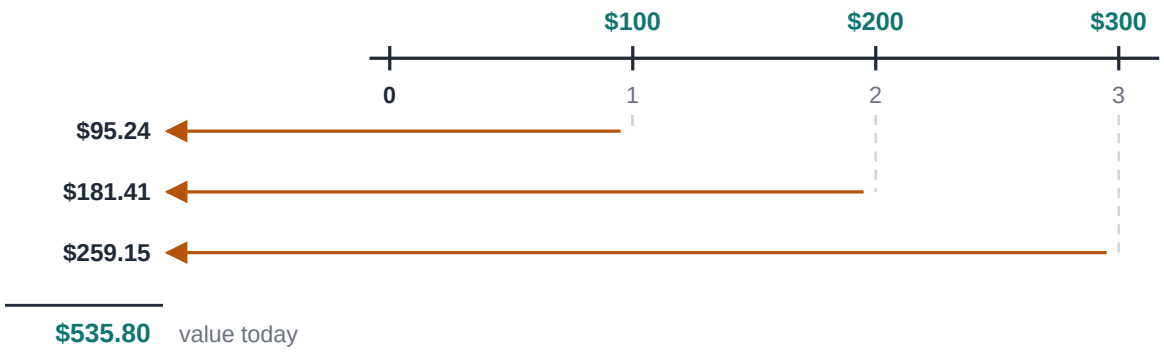


Figure 4. Uneven amounts: bring each one back to today on its own, then add. Here at 5%.

> **Watch out**
>
> - Each amount gets its own exponent: its own distance from today.
> - Adding the raw amounts ($600) ignores time completely and overstates the value.

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6. Shortcuts for payments that follow a pattern

The payments repeat in a pattern. Is there a faster way than discounting each one?

Yes, four of them. Each assumes the payments are evenly spaced, one per period, and the first payment comes **one period after** the date the formula gives you.

Annuity: the same payment for a set number of periods

$$\mathrm{PV} = \frac{C}{r} \times \left[1 - \frac{1}{(1+r)^{N}} \right]$$

Growing annuity: a payment that grows by g each period, for a set number of periods

$$\mathrm{PV} = \frac{C}{r-g} \times \left[1 - \left(\frac{1+g}{1+r} \right)^{N} \right]$$

Perpetuity: the same payment forever

$$\mathrm{PV} = \frac{C}{r}$$

Growing perpetuity: a payment that grows by g each period, forever

$$\text{PV} = \frac{C}{r - g} \quad \text{(only when g is less than r)}$$

In the growing versions, C is the **first** payment. Each later one is (1 + g) times the one before.

Annuity	$1,000 a year for 5 years at 6%: $1,000 / 0.06 × [1 − 1 / 1.06^5] = **$4,212.36**
Growing annuity	First payment $1,000, growing 3% a year, 5 years, at 6%: $1,000 / 0.03 × [1 − (1.03 / 1.06)^5] = **$4,457.43**
Perpetuity	$1,000 a year forever at 5%: $1,000 / 0.05 = **$20,000.00**
Growing perpetuity	First payment $1,000, growing 3% forever, at 6%: $1,000 / (0.06 − 0.03) = **$33,333.33**
Check	Each answer matches discounting every payment one by one (the companion script checks this).

The shortcut lands one period early

Each shortcut gives a value **one period before the first payment**. When the first payment is at period 1, that's today and you're done. When payments start later, use part 4 to move the answer back to today.

$$\text{Value today} = \frac{\text{shortcut answer}}{(1 + r)^{t-1}} \quad \text{(first payment at period t)}$$

Setup	Four payments of $1,000, the first at the end of year 3, at 6%.
Shortcut	$1,000 / 0.06 × [1 − 1 / 1.06^4] = **$3,465.11**, a value at year 2 (not today)
Back to today	$3,465.11 ÷ 1.06^2 = **$3,083.93**

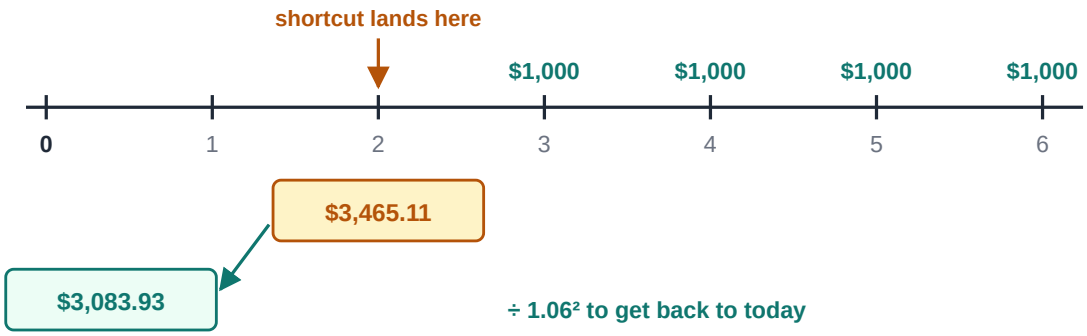


Figure 5. Four payments that start in year 3, at 6%. The annuity formula answers at year 2 (one period before the first payment), so one more step is needed.

Watch out

- The most common mistake: forgetting that the shortcut's answer sits at (first payment − 1), not at today.
- A growing perpetuity only works when g is less than r. If growth is equal to or faster than the rate, the value has no limit and the formula breaks.
- When r equals g in a growing annuity, the formula divides by zero. Each payment is then worth C / (1 + r) today, so the value is N × C / (1 + r).
- Check that the rate, the payment and the growth are all per the same period.

7. Combinations: messier timelines

The payments change partway through. How do I value the whole thing?

Break the timeline into pieces that each match one pattern from parts 2 to 6. Value each piece on its own, move every piece to today, then add. Present values at the same date can always be added.

Example. $1,000 a year for years 1 to 5, then $1,500 a year forever starting in year 6, at 6%.

Piece 1: annuity	$1,000 for 5 years, first payment at year 1, so the answer is already today: **$4,212.36**
Piece 2: perpetuity	$1,500 / 0.06 = $25,000.00. First payment at year 6, so this value sits at year 5
Piece 2 today	$25,000.00 ÷ 1.06^5 = **$18,681.45**
Total today	$4,212.36 + $18,681.45 = **$22,893.82**

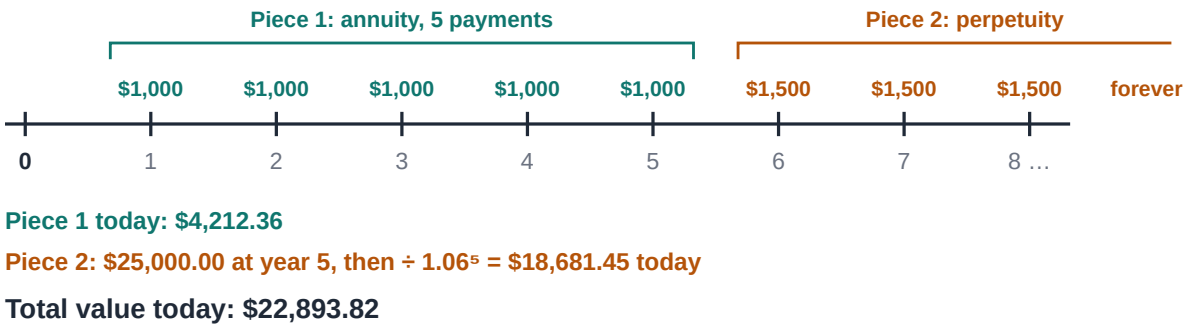


Figure 6. A combination: split the timeline into patterns you know, value each piece today, then add.

Watch out

- Every piece must be moved to the same date before you add.
- Each piece's "one period before" date is set by that piece's own first payment.
- Draw the timeline first (part 1). Most mistakes here are about dates, not arithmetic.

Quick reference

Question	Formula	Part
What will an amount grow to?	$C \times (1 + r)^n$	2
How much was interest on interest?	total interest $- C \times r \times n$	2
What is a future amount worth today?	$C / (1 + r)^n$	3
Move an amount between two dates?	$\times (1 + r)^{gap}$ forward, $\div (1 + r)^{gap}$ back	4
A stream of different amounts?	$\Sigma\, C_t / (1 + r)^t$	5
Same payment, N periods?	$C / r \times [1 - 1 / (1 + r)^N]$	6
Growing payment, N periods?	$C / (r - g) \times [1 - ((1 + g) / (1 + r))^N]$	6
Same payment forever?	C / r	6
Growing payment forever ($g < r$)?	$C / (r - g)$	6
Shortcut when payments start at period t?	shortcut answer $\div (1 + r)^{t - 1}$	6
Payments that change partway?	value each piece today, then add	7

Every number in this guide was generated and cross-checked by the companion script, *time_value.py*.